Exercise 1: Map the Roman Empire

Dataset: Roman Empire Infrastructure

In this exercise you make two maps of the Roman Empire and a layout with both maps. You practise loading raster and vector data, layer order, raster styling, categorized and graduated styles, labels, pie charts, map themes, coordinate systems and layouts.

A) Map the empire and its landscape

1. Load all layers of the dataset. Put them in a logical order.
2. Show `elevation`: low land in green, mountains in brown or red. Use *Identify* to find the elevation of a city in the Alps.
3. Show the provinces with an outline only, so that the elevation stays visible.
4. Show the roads with a categorized style: major roads thicker than minor roads.
5. Show the cities with a categorized style on `type`: cities as small dots, legionary fortresses as squares. Label the cities with their `name`.
6. Save the current styles as a map theme.

B) Map the infrastructure per province

1. Restyle the provinces. Make sure you don't lose your previous style: first create a new style, then start configuring.
2. Colour the provinces with a graduated style on `area_km2`: the larger the province, the darker the colour. Label the provinces with their `name`.
3. Add a pie chart to every province with the number of monuments of each type (`n_theatres`, `n_amphitheatres`, `n_circuses`, `n_baths`, `n_temples`). The more monuments in total (`n_monuments`), the larger the pie chart.
4. Save the current styles as a map theme. Switch between the two map themes to check that both maps are kept.

C) Prepare for finishing

1. Set the project to a projected CRS that suits the empire and compare the shapes of the provinces before and after.
2. Save the project in the folder of the dataset. **Keep it**: you need it in the next step.

D) Make a layout

Make a finished map, as a PDF, with:

1. An **overview map** of the whole empire, with the style of Map 2 (provinces and pie charts). This is the largest map on the page. It shows a frame with the extent of the detail map.
2. A **detail map** that zooms in on the province of Belgica, with the style of Map 1 (elevation, roads and cities).
3. A title, and a legend, a scale bar and a north arrow that are linked to the overview map.